# STAFF REVIEW · H446 194 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

## Do Now — source answers

1. 11111110.

2. 00010000.

3. 0.5.

4. 4.

## I Do — source worked steps

1. First shift — 0.0011 becomes 0.0110; reduce exponent 5 → 4. It still begins 00.

2. Second shift — 0.0110 becomes 0.1100; reduce exponent 4 → 3. The first two bits are now 01.

3. Check — Both represent 6 in decimal, so normalisation preserved the value.

## We Do — guided CHECK

1. Adjust and check
Exponent decreases from 4 to 3. Both representations equal decimal 5.

## You Do — source mark schemes

1. Mantissa $0.1100_2$; exponent 3.

2. $0.0010 \times 2^5 \rightarrow 0.0100 \times 2^4 \rightarrow 0.1000 \times 2^3$.

3. Redundant leading sign bits are removed so available mantissa positions hold more significant information. Exponent adjustment preserves the represented value.

## Exit Ticket — source answer

1. False. To preserve the value, the exponent must decrease to 3.